

Construction of Recursion Formulas for Lax Pairs of (2+1)-Dimensional Equation: Modified Generalized Dispersive Long Wave Equation*

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Abstract In this article, we study the Lax pairs of (2+1)-dimensional equation: the modified generalized dispersive long wave (MGDLW) equation. Based on the well-known binary Darboux transformation, we dig out the recursion formulas of the first part of the Lax pairs. Then by further discussion and doing some revisional work, we make the recursion formulas fit for the second part of Lax pairs. At last, some solutions to the MGDLW equation are worked out by using the recursion formula.

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1 Introduction

As we know, to construct recursion formulas is a powerful way for solving the Lax pairs of an integrable nonlinear evolution equations. In Refs. [1] and [2], we have put forward recursion formulas for solving Lax pairs of KdV equation, sine-Gordon equation, nonlinear Schrödinger equation, and the (2+1)-dimensional extension of classical boussinesq system. The ideas for our work are based on the two points:

(i) A Bäcklund transformation^[3] for Lax pairs should be found.

(ii) Based on the fact that characteristic functions of the Lax pairs will be consumed after one time of Bäcklund transformation, new characteristic functions should be found to make sure that we can use the Bäcklund transformation repeatedly.

In this article we will discuss the Lax pairs of the modified generalized dispersive long wave (MGDLW) equation. The first part of the Lax pairs is in the form of a partial differential equation because the derivatives of two variables emerge, while the first part of the Lax pairs shown in Refs. [1] and [2] is in the form of ordinary differential equation because only the derivatives for x emerge. Thus, we should take some new technology to finish the following work. In Sec. 2, we will give a detailed discussion for solving the Lax pairs of MGDLW equation. Then, in Sec. 3, we will put forward recursion formulas for solving Lax pairs of MGDLW equation and some solutions are digged out by using the recursion formulas.

2 Analysis of Lax Pairs of MGDLW Equation

The MGDLW equation

$$M_t - N_x = 0, \quad M_y^2(N_{yt} - M_{xxy}) + M_{xy}(N_y^2 - M_{xy}^2) + 2M_y(M_{xy}M_{xxy} - N_yN_{xy}) - 4M_y^3M_{xx} = 0, \quad (1)$$

which is a modified version of the generalized dispersive long wave (GDLW) equation^[4,5] $u_{ty} + (\eta_{xy} + 2uu_y)_x = 0$, $\eta_{ty} + (u_{xy} + 2u\eta_y)_x = 0$, is put forward in Ref. [4]. Furthermore, the author^[4] shows a profound relationship between Eqs. (1) and long dispersive wave equation,^[6] which is in the form:

$$(uw)_x = V_y, \quad \lambda u_t + u_{xx} - 2uV = 0, \quad \lambda w_t + w_{xx} + 2wV = 0.$$

Therefore it is meaningful to study MGDLW equation. In Ref. [4], the following Lax pairs for MGDLW equation (1) were obtained,

$$\varphi_t - \varphi_{xx} - 2M_x\varphi = 0, \quad (2a)$$

$$2M_y\varphi_{xy} - (M_{xy} + N_y)\varphi_y + 2M_y^2\varphi = 0. \quad (2b)$$

Notice the invariance in MGDLW equation (1): $\partial t \Rightarrow -\partial t$, $N \Rightarrow -N$, we obtain another Lax pair for Eq. (1),

$$\phi_t + \phi_{xx} + 2M_x\phi = 0, \quad (3a)$$

$$2M_y\phi_{xy} - (M_{xy} - N_y)\phi_y + 2M_y^2\phi = 0. \quad (3b)$$

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In order to construct recursion formulas for Lax pairs (1) and (2), we should review the well-known results called binary Darboux transformation,^[7]

Lemma 2.1

For the following twin system^[7]

$$-\alpha\varphi_t + \varphi_{xx} + u\varphi = 0, \quad (4a)$$

$$\alpha\phi_t + \phi_{xx} + u\phi = 0, \quad (4b)$$

if (φ, u) is the solution to Eq. (4a), while (ϕ, u) and (ϕ_1, u) are the solutions to Eq. (4b), then $(\bar{\varphi}, \bar{u})$ is the solution to Eq. (4a), while $(\bar{\phi}, \bar{u})$ is the solution to Eq. (4b), where

$$\bar{u} = u + 2(\ln \phi_1)_{xx}, \quad \bar{\phi} = \phi_x - \frac{\phi_{1x}}{\phi_1}\phi, \quad \bar{\varphi} = \left(A_1 + B_1 \int_{(x_0, y_0)}^{(x, y)} (\phi_1 \varphi dx + \alpha^{-1}(\phi_1 \varphi_x - \phi_{1x} \varphi) dt) \right) \frac{1}{\phi_1},$$

while A_1 and B_1 are arbitrary constants.

Directly applying Lemma 1 to Eqs. (2a) and (3a), we obtain Lemma 2.2.

Lemma 2.2

If (φ, M) is the solution to Eq. (2a), while (ϕ, M) and (ϕ_1, M) are the solutions to Eq. (3a), then $(\bar{\varphi}, \bar{M})$ is the solution to Eq. (2a), while $(\bar{\phi}, \bar{M})$ is the solution to Eq. (3a), where

$$\bar{M} = M + (\ln \phi_1)_x, \quad \bar{\phi} = \phi_x - \frac{\phi_{1x}}{\phi_1}\phi, \quad \bar{\varphi} = \left(A + \int_{(x_0, y_0)}^{(x, y)} (\phi_1 \varphi dx + (\phi_1 \varphi_x - \phi_{1x} \varphi) dt) \right) \frac{1}{\phi_1},$$

while A is an arbitrary constant.

In the following, we assume that $(\varphi_n^{(1)}, M_n, N_n)$ and $(\varphi_n^{(2)}, M_n, N_n)$ are the known solutions of Lax pairs (2) while $(\phi_n^{(1)}, M_n, N_n)$ and $(\phi_n^{(2)}, M_n, N_n)$ are the known solutions to Eqs. (3). Our aim is to establish the recursion formulas for Lax pairs (2) and (3). In fact, by combination of using formulas shown in Lemma 2.2, we can obtain the recursion formulas for the first part of Lax pairs (2) and (3). Here, for clarification, we just announce our results.

Theorem 2.1

If $(\varphi_n^{(1)}, M_n)$ and $(\varphi_n^{(2)}, M_n)$ are the solutions to Eq. (2a), while $(\phi_n^{(1)}, M_n)$ and $(\phi_n^{(2)}, M_n)$ are the solutions to Eq. (3a), then $(\varphi_{n+1}^{(1)}, M_{n+1})$ and $(\varphi_{n+1}^{(2)}, M_{n+1})$ are the solutions to Eq. (2a), while $(\phi_{n+1}^{(1)}, M_{n+1})$ and $(\phi_{n+1}^{(2)}, M_{n+1})$ are the solutions to Eq. (3a), where

$$\begin{aligned} M_{n+1} &= M_n + \frac{\partial}{\partial x} \left(\ln \left(A_n + \int_{(x_0, y_0)}^{(x, y)} ((\varphi_n^{(2)} \phi_n^{(2)}) dx + (\varphi_{nx}^{(2)} \phi_n^{(2)} - \phi_{nx}^{(2)} \varphi_n^{(2)}) dt) \right) \right), \\ \varphi_{n+1}^{(1)} &= \varphi_n^{(2)} \left(A_n + \int_{(x_0, y_0)}^{(x, y)} ((\varphi_n^{(2)} \phi_n^{(2)}) dx + (\varphi_{nx}^{(2)} \phi_n^{(2)} - \phi_{nx}^{(2)} \varphi_n^{(2)}) dt) \right)^{-1}, \\ \varphi_{n+1}^{(2)} &= \varphi_n^{(1)} - \varphi_n^{(2)} \frac{B_n + \int_{(x_0, y_0)}^{(x, y)} ((\varphi_n^{(1)} \phi_n^{(2)}) dx + (\varphi_{nx}^{(1)} \phi_n^{(2)} - \phi_{nx}^{(2)} \varphi_n^{(1)}) dt)}{A_n + \int_{(x_0, y_0)}^{(x, y)} ((\varphi_n^{(2)} \phi_n^{(2)}) dx + (\varphi_{nx}^{(2)} \phi_n^{(2)} - \phi_{nx}^{(2)} \varphi_n^{(2)}) dt)}, \\ \phi_{n+1}^{(1)} &= \phi_n^{(2)} \left(A_n + \int_{(x_0, y_0)}^{(x, y)} ((\varphi_n^{(2)} \phi_n^{(2)}) dx + (\varphi_{nx}^{(2)} \phi_n^{(2)} - \phi_{nx}^{(2)} \varphi_n^{(2)}) dt) \right)^{-1}, \\ \phi_{n+1}^{(2)} &= \phi_n^{(1)} - \phi_n^{(2)} \frac{C_n + \int_{(x_0, y_0)}^{(x, y)} ((\phi_n^{(1)} \varphi_n^{(2)}) dx + (\varphi_{nx}^{(2)} \phi_n^{(1)} - \phi_{nx}^{(1)} \varphi_n^{(2)}) dt)}{A_n + \int_{(x_0, y_0)}^{(x, y)} ((\varphi_n^{(2)} \phi_n^{(2)}) dx + (\varphi_{nx}^{(2)} \phi_n^{(2)} - \phi_{nx}^{(2)} \varphi_n^{(2)}) dt)}, \end{aligned}$$

while A_n , B_n , and C_n are arbitrary constants.

Proof

Without loss of generality, we only check that $(\varphi_{n+1}^{(2)}, M_{n+1})$ satisfies Eq. (2a), i.e.

$$\varphi_{(n+1)t}^{(2)} - \varphi_{(n+1)xx}^{(2)} - 2M_{(n+1)x} \varphi_{n+1}^{(2)} = 0. \quad (5)$$

In fact, let

$$\Delta_n = B_n + \int_{(x_0, y_0)}^{(x, y)} ((\varphi_n^{(1)} \phi_n^{(2)}) dx + (\varphi_{nx}^{(1)} \phi_n^{(2)} - \phi_{nx}^{(2)} \varphi_n^{(1)}) dt), \quad (6)$$

$$\Omega_n = A_n + \int_{(x_0, y_0)}^{(x, y)} ((\varphi_n^{(2)} \phi_n^{(2)} dx) + (\varphi_{nx}^{(2)} \phi_n^{(2)} - \phi_{nx}^{(2)} \varphi_n^{(1)}) dt), \quad (7)$$

then

$$\varphi_{n+1}^{(2)} = \varphi_n^{(1)} - \varphi_n^{(2)} \Delta_n \Omega_n^{-1}, \quad M_{n+1} = M_n + \Omega_{nx} \Omega_n^{-1}.$$

Thus

$$\begin{aligned} \varphi_{(n+1)t}^{(2)} - \varphi_{(n+1)xx}^{(2)} - 2M_{(n+1)x} \varphi_{n+1}^{(2)} &= (\varphi_{nt}^{(1)} - \varphi_{nxx}^{(1)} - 2M_{nx} \varphi_n^{(1)}) + (\varphi_{nt}^{(2)} - \varphi_{nxx}^{(2)} - 2M_{nx} \varphi_n^{(2)}) \Delta_n \Omega_n^{-1} \\ &+ (-\varphi_n^{(2)} \Delta_{nt} + 2\varphi_{nx}^{(2)} \Delta_{nx} + \varphi_n^{(2)} \Delta_{nxx} - 2\varphi_n^{(1)} \Omega_{nxx}) \Omega_n^{-1} + (-2\varphi_n^{(2)} \Omega_{nx} \Delta_{nx} + 2\varphi_n^{(1)} \Omega_{nx}^2) \Omega_n^{-2} \\ &+ (\varphi_n^{(2)} \Omega_{nxx} + \varphi_n^{(2)} \Omega_{nt} - 2\varphi_{nx}^{(2)} \Omega_{nx}) \Delta_n \Omega_n^{-2}. \end{aligned} \quad (8)$$

By Eqs. (2) and (6) ~ (8), we can check that equation (5) holds. This ends the proof.

By the above discussion, we have digged out the recursion formulas for the first parts of Lax pairs (2) and (3). Next we will do some work to make the formulas fit for the second parts of Lax pairs (2) and (3). Here, for further discussion, we put forward the following remark at first.

Remark 2.1

We emphasis two facts useful here:

(i) For the reason that

$$\bar{M} = M + (\ln \phi_1)_x, \quad M_t = N_x,$$

we can assert that the Bäcklund transformation for N is

$$\bar{N} = N + (\ln \phi_1)_t.$$

(ii) By lemma 2.2, we know that there is a simple recursion relationship from Eq. (3a) to Eq. (2a),

$$\phi_1 \longrightarrow (\phi_1)^{-1}, \quad M \longrightarrow \bar{M} = M + (\ln \phi_1)_x.$$

We can check directly that there is a parallel recursion relationship from the second part of Lax pairs (3) to the second part of Lax pairs (2):

$$(M, N, \phi_1) \rightarrow (\bar{M}, \bar{N}, (\phi_1)^{-1}),$$

which means

$$2\bar{M}_y(\phi_1^{-1})_{xy} - (\bar{M}_{xy} + \bar{N}_y)(\phi_1^{-1})_y + 2\bar{M}_y^2 \phi_1^{-1} = 0, \quad (9)$$

while

$$2M_y \phi_{1xy} - (M_{xy} - N_y) \phi_{1y} + 2M_y^2 \phi_1 = 0.$$

Theorem 2.1 only shows the recursion formulas for the first parts for Lax pairs (2) and (3). In order to establish the corresponding recursion formulas for the second parts of Lax pairs (2) and (3), we should discuss how we can make the $\bar{\varphi}$ shown in Lemma 2.2 fit for y part of Lax pairs (2). In fact, we can revise the arbitrary constant A to a pending function of y : $H(y)$, i.e.,

$$\bar{\varphi} = \left(H(y) + \int_{(x_0, y_0)}^{(x, y)} (\phi_1 \varphi dx + (\phi_1 \varphi_x - \phi_{1x} \varphi)) dt \right) \frac{1}{\phi_1} = \frac{\Delta + H(y)}{\phi_1}, \quad (10)$$

where

$$\Delta = \int_{(x_0, y_0)}^{(x, y)} \phi_1 \varphi dx + (\phi_1 \varphi_x - \phi_{1x} \varphi) dt. \quad (11)$$

In the following, we determine $H(y)$ by substituting Eq. (10) into Eq. (2) and we obtain

$$\begin{aligned} 2\bar{M}_y(\bar{\varphi})_{xy} - (\bar{M}_{xy} + \bar{N}_y)\bar{\varphi}_y + 2\bar{M}_y^2 \bar{\varphi} &= (\Delta + H(y))(2\bar{M}_y(\phi_1^{-1})_{xy} - (\bar{M}_{xy} + \bar{N}_y)(\phi_1^{-1})_y + 2\bar{M}_y^2 \phi_1^{-1}) + 2(M_y + (\ln \phi_1)_{xy}) \\ &\times \left[\left(\Delta_y + \frac{dH}{dy} \right) (\phi_1^{-1})_x + \Delta_x (\phi_1^{-1})_y + \frac{\Delta_{xy}}{\phi_1} \right] - \left[M_{xy} + N_y + (\ln \phi_1)_{xy} + \left(\frac{\phi_{1t}}{\phi_1} \right)_y \right] \left(\Delta_y + \frac{dH}{dy} \frac{1}{\phi_1} \right). \end{aligned}$$

By using Eqs. (3), (9), and (11), we obtain

$$\begin{aligned} M_y [-2M_y \phi_{1x} \phi_1^{-2} + (M_{xy} - N_y) \phi_1^{-1}] \frac{dH}{dy} &+ M_y [-2M_y \phi_{1x} \phi_1^{-2} + (M_{xy} - N_y) \phi_1^{-1}] \Delta_y \\ &+ 2\varphi_y (M_y^2 + M_y \phi_{1xy} \phi_1^{-1} - M_y \phi_{1x} \phi_{1y} \phi_1^{-2}) = 0. \end{aligned} \quad (12)$$

By using Eq. (3) and letting

$$K = -2M_y\phi_{1x}\phi_1^{-2} + (M_{xy} - N_y)\phi_1^{-1}, \quad (13)$$

equation (12) reduces to

$$KM_y \frac{dH}{dy} + KM_y\Delta_y + K\varphi_y\phi_{1y} = 0. \quad (14)$$

In order to determine $H(y)$, we consider the above equation (14) in three cases.

(i) $M_y \neq 0$

By Eq. (14), we can let $H(y)$ satisfy

$$M_y \frac{dH}{dy} + M_y\Delta_y + \varphi_y\phi_{1y} = 0,$$

which means

$$\frac{dH}{dy} = -\Delta_y - M_y^{-1}\varphi_y\phi_{1y}. \quad (15)$$

By Eqs. (2b) and (3b), we can check directly that the right-hand side of Eq. (15) is the function of y only, i.e.

$$(-\Delta_y - M_y^{-1}\varphi_y\phi_{1y})_x = 0, \quad (-\Delta_y - M_y^{-1}\varphi_y\phi_{1y})_t = 0.$$

Thus, doing the integration for y in Eq. (15), we obtain $H(y)$ expressed as

$$H(y) = -\Delta - \int_{y_0}^y M_y^{-1}\varphi_y\phi_{1y} dy. \quad (16)$$

(ii) $M_y = 0, N_y \neq 0$

By Eqs. (2) and (3), we know that

$$\varphi_y = 0, \quad \phi_{1y} = 0,$$

which means that equation (14) absolutely holds. Thus we can take $H(y)$ as an arbitrary function of y .

(iii) $M_y = 0, N_y = 0$

By Eq. (13), we know that $K = 0$, which also means equation (14) absolutely holds. Thus we can also take $H(y)$ as an arbitrary function of y .

In conclusion, we can take $H(y)$ expressed by Eq. (16) under the condition of $M_y \neq 0$ and can take $H(y)$ as an arbitrary function of y under the condition of $M_y = 0$.

3 Conclusions and Solutions to MGDWL Equation

In the following, we will draw a conclusion. In order to state our theory clearly, we define

$$\begin{aligned} \Delta_n &= \int_{(x_0, y_0)}^{(x, y)} \varphi_n^{(1)} \phi_n^{(2)} dx + (\phi_n^{(2)} \varphi_{nx}^{(1)} - \phi_{nx}^{(2)} \varphi_n^{(1)}) dt, & \delta_n &= \int_{(x_0, y_0)}^{(x, y)} \phi_n^{(1)} \varphi_n^{(2)} dx + (\varphi_{nx}^{(2)} \phi_n^{(1)} - \phi_{nx}^{(1)} \varphi_n^{(2)}) dt, \\ \Omega_n &= \int_{(x_0, y_0)}^{(x, y)} \varphi_n^{(2)} \phi_n^{(2)} dx + (\phi_n^{(2)} \varphi_{nx}^{(2)} - \phi_{nx}^{(2)} \varphi_n^{(2)}) dt, \\ H_{\Delta_n} &= -\Delta_n - \int_{y_0}^y M_{ny}^{-1} \varphi_{ny}^{(1)} \phi_{ny}^{(2)} dy, & H_{\delta_n} &= -\delta_n - \int_{y_0}^y M_{ny}^{-1} \varphi_{ny}^{(2)} \phi_{ny}^{(1)} dy, & H_{\Omega_n} &= -\Omega_n - \int_{y_0}^y M_{ny}^{-1} \varphi_{ny}^{(2)} \phi_{ny}^{(2)} dy. \end{aligned} \quad (17)$$

According to discussions in Sec. 2, we can conclude the following theorem.

Theorem 3.1

If $(\varphi_n^{(1)}, M_n, N_n)$ and $(\varphi_n^{(2)}, M_n, N_n)$ are the solutions of Lax pairs (2), while $(\phi_n^{(1)}, M_n, N_n)$ and $(\phi_n^{(2)}, M_n, N_n)$ are the solutions of Lax pairs (3), then $(\varphi_{n+1}^{(1)}, M_{n+1}, N_{n+1})$ and $(\varphi_{n+1}^{(2)}, M_{n+1}, N_{n+1})$ are the solutions of Lax pairs (2), while $(\phi_{n+1}^{(1)}, M_{n+1}, N_{n+1})$ and $(\phi_{n+1}^{(2)}, M_{n+1}, N_{n+1})$ are the solutions of Lax pairs (3), where

$$\begin{aligned} M_{n+1} &= M_n + \frac{\partial}{\partial x} \left(\ln(H_{\Omega_n}(y) + \int_{(x_0, y_0)}^{(x, y)} (\varphi_n^{(2)} \phi_n^{(2)} dx + (\varphi_{nx}^{(2)} \phi_n^{(2)} - \phi_{nx}^{(2)} \varphi_n^{(2)}) dt) \right), \\ N_{n+1} &= N_n + \frac{\partial}{\partial t} \left(\ln(H_{\Omega_n}(y) + \int_{(x_0, y_0)}^{(x, y)} (\varphi_n^{(2)} \phi_n^{(2)} dx + (\varphi_{nx}^{(2)} \phi_n^{(2)} - \phi_{nx}^{(2)} \varphi_n^{(2)}) dt) \right), \\ \varphi_{n+1}^{(1)} &= \varphi_n^{(2)} \left(H_{\Omega_n}(y) + \int_{(x_0, y_0)}^{(x, y)} (\varphi_n^{(2)} \phi_n^{(2)} dx + (\varphi_{nx}^{(2)} \phi_n^{(2)} - \phi_{nx}^{(2)} \varphi_n^{(2)}) dt \right)^{-1}, \end{aligned}$$

$$\begin{aligned}\varphi_{n+1}^{(2)} &= \varphi_n^{(1)} - \varphi_n^{(2)} \frac{H_{\Delta_n}(y) + \int_{(x_0, y_0)}^{(x, y)} (\varphi_n^{(1)} \phi_n^{(2)} dx + (\varphi_n^{(1)} \phi_{nx}^{(2)} - \phi_{nx}^{(2)} \varphi_n^{(1)}) dt)}{H_{\Omega_n}(y) + \int_{(x_0, y_0)}^{(x, y)} (\varphi_n^{(2)} \phi_n^{(2)} dx + (\varphi_n^{(2)} \phi_{nx}^{(2)} - \phi_{nx}^{(2)} \varphi_n^{(2)}) dt)}, \\ \phi_{n+1}^{(1)} &= \phi_n^{(2)} \left(H_{\Omega_n}(y) + \int_{(x_0, y_0)}^{(x, y)} (\varphi_n^{(2)} \phi_n^{(2)} dx + (\varphi_{nx}^{(2)} \phi_n^{(2)} - \phi_{nx}^{(2)} \varphi_n^{(2)}) dt) \right)^{-1}, \\ \phi_{n+1}^{(2)} &= \phi_n^{(1)} - \phi_n^{(2)} \frac{(H_{\delta_n}(y) + \int_{(x_0, y_0)}^{(x, y)} (\varphi_n^{(2)} \phi_n^{(1)} dx + (\phi_n^{(1)} \varphi_{nx}^{(2)} - \phi_{nx}^{(1)} \varphi_n^{(2)}) dt)}{(H_{\Omega_n}(y) + \int_{(x_0, y_0)}^{(x, y)} (\varphi_n^{(2)} \phi_n^{(2)} dx + (\phi_n^{(2)} \varphi_{nx}^{(2)} - \phi_{nx}^{(2)} \varphi_n^{(2)}) dt)}.\end{aligned}$$

Here, $H_{\Delta_n}(y)$, $H_{\delta_n}(y)$, and $H_{\Omega_n}(y)$ are determined according to two cases.

- (i) When $M_y \neq 0$, $H_{\Delta_n}(y)$, $H_{\delta_n}(y)$, and $H_{\Omega_n}(y)$ are determined by formula (18).
- (ii) When $M_y = 0$, $H_{\Delta_n}(y)$, $H_{\delta_n}(y)$, and $H_{\Omega_n}(y)$ can be taken as arbitrary functions of y .

Example 3.1

We take the solutions of Lax pairs (2): $(\varphi_0^{(1)}, M_0, N_0) = (t + x^2/2, 0, 0)$, $(\varphi_0^{(2)}, M_0, N_0) = (1, 0, 0)$, and the solutions of Lax pairs (3): $(\phi_0^{(1)}, M_0, N_0) = (y + x, 0, 0)$, $(\phi_0^{(2)}, M_0, N_0) = (y, 0, 0)$, then by formula (17) and Theorem 3.1, we have

$$\Delta_0 = \frac{x^3 y}{6} + txy, \quad \delta_0 = yx + \frac{x^2}{2} - t, \quad \Omega_0 = xy.$$

For $M_0 = 0$, we can take $H_{\Delta_0}(y)$, $H_{\delta_0}(y)$, and $H_{\Omega_0}(y)$ as arbitrary functions of y . Therefore

$$\begin{aligned}M_1 &= (\ln(xy + A_0(y)))_x, & N_1 &= (\ln(xy + A_0(y)))_t, \\ \varphi_1^{(1)} &= \frac{1}{xy + A_0(y)}, & \varphi_1^{(2)} &= t + \frac{x^2}{2} - \frac{x^3 y/6 + txy + B_0(y)}{xy + A_0(y)}, \\ \phi_1^{(1)} &= \frac{y}{xy + A_0(y)}, & \phi_1^{(2)} &= y + x - y \frac{yx + x^2/2 - t + C_0(y)}{xy + A_0(y)},\end{aligned}$$

where $A_0(y)$, $B_0(y)$, and $C_0(y)$ are arbitrary functions of y . Especially, if we take $A_0(y) = 0$, $B_0(y) = C_0(y) = e^y$, $\varphi_1^{(1)}$, $\varphi_1^{(2)}$, $\phi_1^{(1)}$ and $\phi_1^{(2)}$ reduce to

$$\varphi_1^{*(1)} = \frac{1}{xy}, \quad \varphi_1^{*(2)} = \frac{x^2}{3} - \frac{e^y}{xy}, \quad \phi_1^{*(1)} = \frac{y}{xy + e^y}, \quad \phi_1^{*(2)} = \frac{x}{2} + \frac{t}{x} - \frac{e^y}{x}.$$

Correspondingly

$$\Delta_1^* = \frac{x}{2y} - \frac{e^y}{xy} - \frac{t}{xy}, \quad \delta_1^* = \frac{x^2}{6} + \frac{e^y}{xy} + t, \quad \Omega_1^* = \frac{x^4}{24} - \frac{x^2 e^y}{6} - \frac{x e^y}{2y} - \frac{e^{2y}}{xy} + \frac{x^2 t}{6} + \frac{t e^y}{xy} + \frac{t^2}{2} - e^y t.$$

Then

$$\begin{aligned}M_2 &= \frac{1}{x} + \frac{\partial}{\partial x} \left(\frac{x^4}{24} - \frac{x^2 e^y}{6} - \frac{x e^y}{2y} - \frac{e^{2y}}{xy} + \frac{x^2 t}{6} + \frac{t e^y}{xy} + \frac{t^2}{2} - e^y t \right), \\ N_2 &= \frac{\partial}{\partial t} \left(\frac{x^4}{24} - \frac{x^2 e^y}{6} - \frac{x e^y}{2y} - \frac{e^{2y}}{xy} + \frac{x^2 t}{6} + \frac{t e^y}{xy} + \frac{t^2}{2} - e^y t \right), \\ \varphi_2^{*(1)} &= \frac{\varphi_1^{*(1)}}{\Omega_1^* + A_1(y)}, & \varphi_2^{*(2)} &= \varphi_1^{*(1)} - \varphi_1^{*(2)} \frac{\Delta_1^* + B_1(y)}{\Omega_1^* + A_1(y)}, \\ \phi_2^{*(1)} &= \frac{\phi_1^{*(2)}}{\Omega_1^* + A_1(y)}, & \phi_2^{*(2)} &= \varphi_1^{*(1)} - \varphi_1^{*(2)} \frac{\Delta_1^* + B_1(y)}{\Omega_1^* + A_1(y)},\end{aligned}$$

where $A_1(y)$, $B_1(y)$, and $C_1(y)$ are arbitrary functions of y .

Example 3.2

We take the solutions of Lax pairs (2)

$$(\varphi_0^{(1)}, M_0, N_0) = \left(\exp\left(a_1 x - \frac{ky}{a_1} + a_1^2 t\right), ky, 0 \right), \quad (\varphi_0^{(2)}, M_0, N_0) = \left(\exp\left(a_2 x - \frac{ky}{a_2} + a_2^2 t\right), ky, 0 \right),$$

and the solutions of Lax pairs (3)

$$(\phi_0^{(1)}, M_0, N_0) = \left(\exp\left(c_1 x - \frac{ky}{c_1} - c_1^2 t\right), ky, 0 \right), \quad (\phi_0^{(2)}, M_0, N_0) = \left(\exp\left(c_2 x - \frac{ky}{c_2} - c_2^2 t\right), ky, 0 \right),$$

where a_i , c_i , and k are arbitrary constants. Then by letting

$$\theta_i = a_i x - \frac{ky}{a_i} + a_i^2 t, \quad \rho_i = c_i x - \frac{ky}{c_i} + c_i^2 t \quad (i = 1, 2),$$

and using formulas (17) and (18), we have

$$\begin{aligned} \Delta_0 &= \frac{\exp(\theta_1 + \rho_2)}{a_1 + c_2}, & \delta_0 &= \frac{\exp(\theta_2 + \rho_1)}{a_2 + c_1}, & \Omega_0 &= \frac{\exp(\theta_2 + \rho_2)}{a_2 + c_2}, \\ H_{\Delta_0}(y) &= \beta_1, & H_{\delta_0}(y) &= \beta_2, & H_{\Omega_0}(y) &= \beta_3. \end{aligned}$$

Here β_j can be taken as arbitrary constant ($j = 1, 2, 3$). By theorem 3.1

$$\begin{aligned} M_1 &= ky + \frac{\partial}{\partial x} \left[\ln \left(\exp \left((a_2 + c_2)x - \left(\frac{k}{a_2} + \frac{k}{c_2} \right) y + (a_2^2 - c_2^2)t \right) + \beta_3 \right) \right], \\ N_1 &= \frac{\partial}{\partial t} \left[\ln \left(\exp \left((a_2 + c_2)x - \left(\frac{k}{a_2} + \frac{k}{c_2} \right) y + (a_2^2 - c_2^2)t \right) + \beta_3 \right) \right], \\ \varphi_1^{(1)} &= \frac{\exp(\theta_2)}{\exp(\theta_2 + \rho_2) + \beta_3}, \\ \varphi_1^{(2)} &= \exp \theta_1 - \frac{(a_2 + c_2)(\exp(\theta_1 + \theta_2 + \rho_2) + \beta_1 \exp(\theta_2))}{(a_1 + c_2)(\exp(\theta_2 + \rho_2) + \beta_3)}, \\ \phi_1^{(1)} &= \frac{\exp(\rho_2)}{\exp(\theta_2 + \rho_2) + \beta_3}, \\ \phi_1^{(2)} &= \exp \rho_1 - \frac{(a_2 + c_2)(\exp(\theta_2 + \rho_1 + \rho_2) + \beta_2 \exp(\rho_2))}{(a_2 + c_1)(\exp(\theta_2 + \rho_2) + \beta_3)}. \end{aligned}$$

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